

Software Requirements Specification

Yhar Bid — Adaptive Routine & Reminder Assistant

Introduction to Software Engineering (15031001) · Week 3 · Requirements & Specification

Field	Yhar Bid
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Team name	Yhar Bid
Project title	Yhar Bid — Adaptive Routine & Reminder Assistant
SDLC chosen in M1 + reason	Agile — develop the MVP in small iterations, test with real users, collect feedback, and continuously improve the application and behavior strategies.
Date / version	v1.1 · Week 3 SRS revision
Team members	Peerapap Kummongkon; Achitapon Wiangmoon; Nattakul Pasoda; Phakpoom Jaiwoot; Somchay Aryi
Source basis	M1-Charter_YharBid.docx, the team-provided revised requirements in this revision, and the Week 3 SRS worked-example structure.

1. Introduction

1.1 Purpose and scope

Many users know what they need to do but may procrastinate, miss planned activities, or repeatedly ignore generic reminders. Yhar Bid is designed as a role-based AI assistant that combines routine and habit tracking with Google Calendar data so that the system can prepare useful information before important activities and adapt short notifications to the user's preferred communication style.

The revised MVP focuses on six connected capabilities: role-based onboarding and AI style selection, two-way Google Calendar synchronization, proactive role-based task preparation, role-contextual habit tracking, customized short notifications, and a behavioral analytics dashboard. The system supports three role contexts: Student, Corporate Employee, and Educator.

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
Student	Primary user	Connect a calendar, manage habits, receive deadline preparation and short reminders.	Irrelevant preparation or repetitive reminders.
Corporate Employee	Primary user	Connect a calendar, prepare for meetings, manage habits, and receive concise reminders.	Meeting preparation that is not relevant to the scheduled context.
Educator	Primary user	Connect a calendar, prepare teaching/exam content, manage habits, and receive concise reminders.	Generic preparation that ignores the teaching/exam context.
Yar Oo; Achitapon Wiangmoon; Nattakul Pasoda	Tester / stakeholder representatives	Test role selection, calendar synchronization, reminders, habit tracking, proactive preparation, and analytics.	A system that does not reflect actual user behavior or role needs.

Name / role	Kind	What they need	What they fear / forbid
Google Calendar API	External system	Authorized read/write access to calendar events used by synchronization and proactive preparation.	Unauthorized access or incorrect synchronization.
AI Provider	External system	Provide AI-generated preparation, behavior analysis, and personalized notification content through the planned AI service integration.	Service outage, invalid requests, or misuse of the AI service.

2. Overall description

2.1 Product context — what it does / does not

Yhar Bid acts as a role-aware assistant around the user's routines and calendar. It receives authorized Google Calendar information, tracks role-contextual habits, prepares role-specific material for upcoming events, and sends short notifications using the selected AI communication style. The analytics dashboard summarizes adherence and habit success so that the user can understand progress over time.

IN SCOPE (MVP)	OUT OF SCOPE (explicit promise)
FR-1 Role-based onboarding & style selection	Social messaging or group chat inside the app
FR-2 Two-way Google Calendar synchronization	Payment, subscriptions, or monetization features
FR-3 Proactive role-based task preparation	Replacing Google Calendar as a full general-purpose calendar application
FR-4 Role-contextual habit tracker	Automatic changes to user role or preferences without user control
FR-5 Customized short notifications	Unsupported roles outside Student, Corporate Employee, and Educator
FR-6 Behavioral analytics dashboard	Features unrelated to the stated role, calendar, habit, notification, and analytics flows

2.2 Assumptions and constraints

- Assumption: users provide their role, preferred AI communication style, habit goals, and calendar authorization explicitly.
- Constraint: supported role contexts are Student, Corporate Employee, and Educator.
- Constraint: Google Calendar synchronization uses Google OAuth 2.0 and the Google Calendar REST APIs with the user's permission for read/write access.
- Constraint: the planned application stack is Flutter (Dart) with `table_calendar` and `flutter_local_notifications` on the frontend.
- Constraint: Firebase Auth, Firestore, and Cloud Functions are the planned backend/cloud components; Google Gemini API is the planned generative AI service.
- Constraint: background synchronization and proactive preparation use a Background Service or Cloud Functions with a Cron Job/PubSub scheduling mechanism.
- Constraint: AI-generated output must have a short-notification fallback template when the AI service is unavailable.
- Constraint: source code must not hardcode secret keys or OAuth client secrets; secrets must be supplied through `.env/environment` variables or Firebase Secret Manager.

3. User Requirements (Functional Requirements)

Format: each requirement is expressed as a user story with observable acceptance criteria. The six functional requirements below represent the revised MVP described by the team. FR-3 uses synchronized calendar context and the selected role to prepare content, while FR-5 and FR-6 consume user and behavior information produced by the earlier flows.

FR-1 — Role-Based Onboarding & Style Selection

User story: As a new user, I want to select my primary role and preferred AI communication style, so that Yhar Bid can personalize its behavior to my context and communication preference.

Pain / rationale: Different roles have different responsibilities, and users may prefer different tones such as Polite Jarvis, Friendly, or Aggressive Motivator.

1. Given a new user completes registration, when the user selects Student, Corporate Employee, or Educator, then the selected role is stored as the active role context.
2. Given a user selects an AI communication style, when an AI-generated message is produced later, then the stored style is available to the messaging flow.

FR-2 — Google Calendar Synchronization

User story: As a user, I want to connect and synchronize Google Calendar with Yhar Bid, so that calendar activities, deadlines, and meetings can be used by the assistant without duplicate manual entry.

Pain / rationale: Upcoming responsibilities are already represented as calendar events, so the assistant needs authorized access to use the same information.

1. Given the user grants Google Calendar access through Google OAuth 2.0, when synchronization runs, then authorized calendar events are available in Yhar Bid.
2. Given an authorized event is created or updated in Yhar Bid, when synchronization occurs, then the corresponding Google Calendar event is updated; changes from Google Calendar are likewise reflected in Yhar Bid.

FR-3 — Proactive Role-Based Task Preparation

User story: As a user, I want Yhar Bid to prepare role-specific information before important calendar events, so that I can start the task, meeting, teaching activity, or exam with useful preparation already available.

Pain / rationale: The same calendar event can require different preparation depending on whether the user is a Student, Corporate Employee, or Educator.

1. Given a Student has a calendar event representing a deadline at least 3 days ahead, when the proactive preparation process detects it, then Yhar Bid prepares a short task outline before the deadline.
2. Given a Corporate Employee has a scheduled meeting, when the proactive preparation process detects it, then Yhar Bid prepares meeting agenda or discussion points before the meeting starts.
3. Given an Educator has a scheduled teaching session or exam, when the proactive preparation process detects it, then Yhar Bid prepares a corresponding teaching or exam content structure before the scheduled activity.

FR-4 — Role-Contextual Habit Tracker

User story: As a user, I want my habit goals and routine tracking to reflect my selected role, so that habit activities remain relevant to my daily responsibilities.

Pain / rationale: Habit goals are more meaningful when their context matches the user's role instead of using one generic productivity model for everyone.

1. Given a user has selected a role, when habit goals are created or presented, then the system applies the selected role as the active habit context.
2. Given a user records habit outcomes, when the data is stored, then the record can be used to calculate habit success information for the analytics dashboard.

FR-5 — Customized Short Notifications

User story: As a user, I want short notifications that match my selected communication style, so that reminders are concise and more suitable to how I prefer to be addressed.

Pain / rationale: Generic reminders can become ineffective when they repeat the same wording regardless of user preference or context.

1. Given a user has selected a communication style, when a notification is generated, then the message is short, direct, and consistent with the selected style.
2. Given the AI generation service is unavailable, when a notification is needed, then the system sends an available fallback short-notification template instead of failing the reminder flow.

FR-6 — Behavioral Analytics Dashboard

User story: As a user, I want a weekly dashboard showing my calendar adherence and habit success rate, so that I can understand how consistently I follow my planned activities.

Pain / rationale: Users need a consolidated view of whether planned calendar activities and habits are being completed successfully.

1. Given calendar and habit outcome data exists, when the user opens the dashboard, then the system shows a weekly calendar-adherence summary and habit success rate.
2. Given new outcomes are recorded, when the dashboard is refreshed, then the displayed weekly metrics reflect the updated stored data.

4. Engineering & Architecture Requirements (Non-Functional)

The supplied revised project context specifies architecture, technology, security, integration, fallback, error-handling, and coding constraints but does not provide numeric performance thresholds. These requirements therefore preserve the specified engineering constraints without inventing additional numeric targets.

ID	Kind	Statement (testable)	How we will check
NFR-1	Frontend technology	The mobile client shall use Flutter (Dart), with table_calendar for calendar UI and flutter_local_notifications for local notifications.	Inspect project dependencies and frontend module structure.
NFR-2	Backend & cloud	The planned backend shall use Firebase Auth, Firestore, and Cloud Functions, with Google Gemini API for generative AI.	Verify configured services and module integrations in the project.
NFR-3	External integration	Google Calendar access shall use Google OAuth 2.0 and Google REST APIs with authorized read/write calendar scopes for synchronization.	Complete an OAuth authorization flow and verify authorized read/write synchronization.
NFR-4	Architecture	The Flutter codebase shall follow Clean Architecture with clear separation of views/, controllers/, services/, and models/, including google_calendar_service.dart and ai_service.dart where applicable.	Inspect repository structure and confirm responsibility boundaries.

ID	Kind	Statement (testable)	How we will check
NFR-5	Authentication & security	OAuth Client Secrets and other secret keys shall not be hardcoded in source code and shall be supplied through environment variables (.env) or Firebase Secret Manager.	Search the repository for hardcoded secrets and inspect the configured secret source.
NFR-6	Background sync & proactive scheduler	A background service or Cloud Functions using Cron Job/PubSub shall periodically scan Google Calendar events and trigger proactive preparation before relevant activities.	Create test calendar events and verify that the scheduled preparation flow is triggered.
NFR-7	AI fallback & availability	When the AI API is unavailable, the system shall continue the core reminder flow by using a fallback short-notification template.	Disable or mock the AI API and verify that a notification still reaches the user.
NFR-8	Error handling	Application services shall use custom exceptions and return the defined standard JSON error shape: {"status": false, "message": "...", "data": null}.	Trigger representative service errors and verify the exception mapping and response schema.
NFR-9	Coding style	Code shall use strong typing (type hints) and explicit return types for functions, with Docstrings/Comments for logic; role-based prompting logic shall be documented in Thai.	Review source files against the team coding standard.
NFR-10	Module scope discipline	Developers shall not modify files outside the feature/module assigned in the project issue.	Review pull requests or commits for unrelated file changes.

5. Use cases

5.1 Use-case list

Use case	Name	Actor	Related FR	Brief success story
UC-1	Role-Based Onboarding	User	FR-1	User registers and selects Student, Corporate Employee, or Educator as the active role.
UC-2	Select AI Communication Style	User	FR-1 / FR-5	User selects a preferred communication style for AI-generated messages.
UC-3	Google Calendar Synchronization	User / Google Calendar API	FR-2	User authorizes access and calendar data is synchronized in both directions.
UC-4	Proactive Role-Based Task Preparation	Yharbid AI Assistant / AI Provider	FR-3	System detects relevant calendar events and prepares role-specific content.
UC-5	Prepare Student Task Outline	Yharbid AI Assistant / AI Provider → User	FR-3	For a Student deadline, the system prepares a short outline ahead of the deadline.
UC-6	Prepare Employee Meeting Agenda	Yharbid AI Assistant / AI Provider → User	FR-3	For a Corporate Employee meeting, the system prepares an agenda or discussion points.
UC-7	Prepare Educator Teaching / Exam Structure	Yharbid AI Assistant / AI Provider → User	FR-3	For an Educator class or exam, the system prepares a content structure.
UC-8	Manage Role-Contextual Habits	User	FR-4	User manages habits whose goal context follows the selected role.

Use case	Name	Actor	Related FR	Brief success story
UC-9	Receive Customized Short Notification	Yharbid AI Assistant / AI Provider → User	FR-5	User receives a concise reminder using the selected communication style or a fallback template.
UC-10	View Behavioral Analytics Dashboard	User	FR-6	User views weekly calendar adherence and habit success rate.

5.2 Use-case diagram

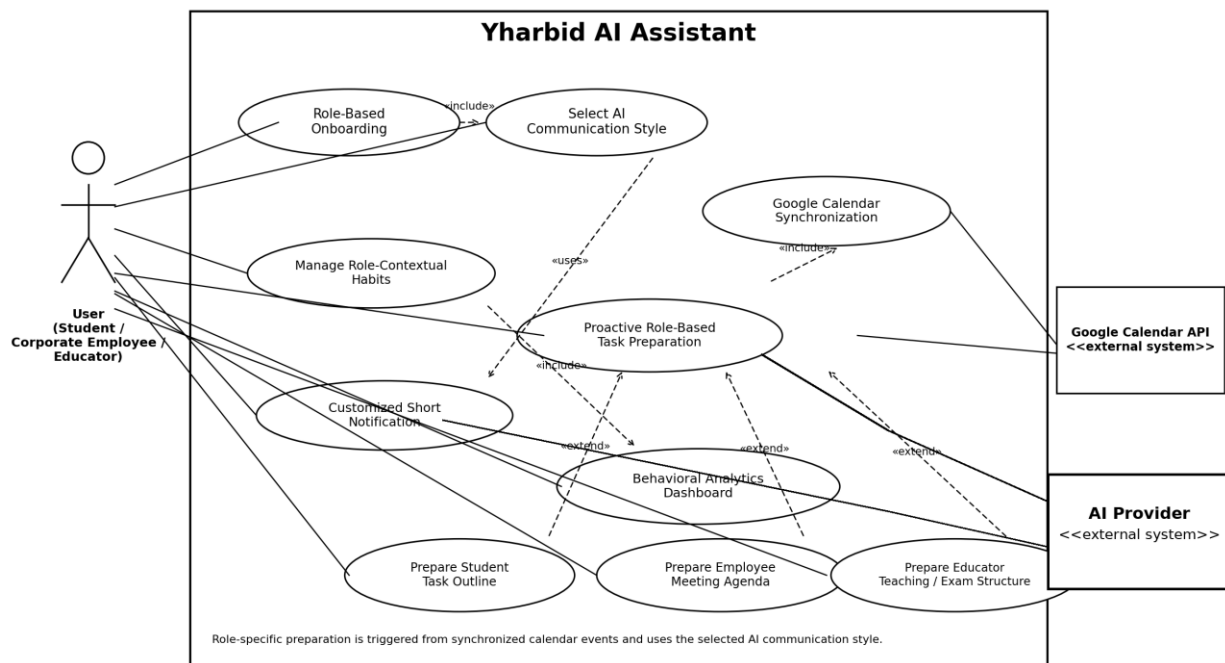


Figure 1. Yharbid use-case diagram aligned with the revised functional requirements and external Google Calendar and AI Provider integrations.

Note: “Prepare Student Task Outline,” “Prepare Employee Meeting Agenda,” and “Prepare Educator Teaching / Exam Structure” are role-specific extensions of the proactive preparation flow. They are not additional functional requirements beyond FR-3.

Note: Yharbid AI Assistant is the system under specification; Google Calendar API and AI Provider are external systems. The AI Provider supports AI-generated preparation, behavior analysis, and personalized notification content.

6. Out of scope

- Social messaging or group chat inside the app.
- Payment, subscription, or monetization features.
- Replacing Google Calendar as a complete general-purpose calendar application.
- Automatic changes to the user’s role or communication preferences without user control.
- Support for roles outside Student, Corporate Employee, and Educator in the MVP.

- Features unrelated to calendar synchronization, role-based preparation, habit tracking, notifications, or analytics.

7. Traceability matrix (Golden Thread)

The revised thread is: different role contexts and upcoming calendar events → synchronized schedule and habit data → proactive role-specific preparation and concise notifications → weekly analytics feedback. FR-1 defines context and communication preference; FR-2 provides calendar input; FR-4 supplies habit outcomes; FR-5 and FR-6 turn those inputs into user-facing adaptation and measurement.

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
Different roles require different preparation.	FR-1 Role + style selection	Role context + communication style	Role-based onboarding / preference flow	Role selection tests
Calendar information is already part of users' schedules.	FR-2 Calendar sync	Google Calendar OAuth + read/write synchronization	Calendar integration service	Two-way sync tests
Upcoming deadlines, meetings, teaching, and exams need preparation.	FR-3 Proactive preparation	Role-based preparation pipeline	Outline / agenda / structure generation	Role-specific preparation tests
Users need habit goals relevant to their role.	FR-4 Role-contextual habits	Role-aware habit data model and tracking	Habit tracker	Habit context and persistence tests
Generic reminders may be ignored.	FR-5 Short notifications	Style-aware message strategy + fallback template	Notification feature	Style and fallback tests
Users need a consolidated view of adherence and habit success.	FR-6 Analytics dashboard	Weekly aggregation of calendar and habit outcomes	Behavioral analytics dashboard	Dashboard calculation tests

8. AI usage log

This SRS includes AI-assisted drafting. The table below records AI-assisted work used during the development of the document; the team should verify requirements against stakeholder feedback before submission.

Date	Tool / model	What we asked	What AI produced	What we changed / verified	What we learned
10 Sep 2026	ChatGPT	Draft project-specific functional requirements from the Yhar Bid M1 charter.	Drafts for FR-1 to FR-5 covering routines, reminders, behavior records, analysis, and personalized messages.	Team should compare each draft with M1 wording and confirm the final acceptance criteria.	Requirements are stronger when each FR has a clear user outcome and testable condition.
12 Sep 2026	ChatGPT	Draft measurable non-functional requirements and use cases for the SRS.	Proposed performance, notification reliability, usability, privacy, reliability, and AI fallback requirements plus five use cases.	Team should confirm numeric thresholds and test methods with actual stakeholders before submission.	NFRs need measurable targets rather than vague words such as "fast" or "easy".
13 Sep 2026	ChatGPT	Organize the SRS into scope, traceability, and AI usage log sections using the Week 3 example structure.	Structured the five FRs into a problem → requirement → solution → feature → test thread and aligned the use-case list.	Removed StudyMate-specific content and kept only the document structure relevant to Yhar Bid.	The SRS should preserve Yhar Bid's own problem and MVP while using the example's organization.
5 Oct 2026	ChatGPT	Update the SRS functional requirements to match the revised Yharbid context: role-based onboarding, AI	Reworked the functional-requirements section into role-based onboarding and style selection, Google	Checked that the revised features remain connected to the intended Yharbid AI Assistant scope and	Requirements are clearer when user roles, external integrations, and proactive AI

Date	Tool / model	What we asked	What AI produced	What we changed / verified	What we learned
		communication style selection, Google Calendar synchronization, proactive role-based task preparation, habit tracking, short notifications, and analytics.	Calendar synchronization, proactive preparation for Student/Corporate Employee/Educator roles, role-contextual habit tracking, customized short notifications, and the behavioral analytics dashboard.	separated the external Google Calendar integration from internal application behavior.	actions are stated explicitly.
5 Oct 2026	ChatGPT	Add an external AI Provider actor to the use-case model and align the actors with the revised architecture.	Updated the use-case model to include AI Provider as an external system actor and connected it to AI-related use cases such as task preparation, behavior analysis, and personalized message generation.	Reviewed actor names and relationships so AI Provider is external to Yharbid and Google Calendar API remains a separate external system.	External dependencies should be represented as actors when they participate directly in system interactions.
5 Oct 2026	ChatGPT	Update the engineering and architecture requirements to reflect Flutter, Firebase, Google Gemini, Google Calendar API, OAuth 2.0, background scheduling, fallback notifications, error handling, and coding standards.	Revised the non-functional/engineering requirements to specify the proposed technology stack, Clean Architecture structure, authentication and secret-management rules, background synchronization, AI fallback behavior, standard error responses, and code-quality conventions.	Checked each technology and architecture item against the revised project context supplied by the team and kept the requirements testable where possible.	Architecture requirements are easier to verify when implementation technologies, security rules, and fallback behavior are stated as explicit constraints.

9. Self-check

- ☑ Every FR is written as a user story with a stated user benefit.
- ☑ Every FR has observable acceptance criteria covering the revised role, calendar, preparation, habit, notification, or analytics behavior.
- ☑ FR-3 is the proactive preparation requirement and is explicitly tied to Google Calendar context and the selected user role.
- ☑ Scope and out-of-scope items are stated for the revised MVP.
- ☑ The use-case list covers all six functional requirements and the diagram distinguishes the system boundary from the external Google Calendar API and AI Provider.
- ☑ The Golden Thread traces role/context and calendar inputs through preparation, notifications, and analytics.
- ☑ Engineering and architecture constraints include Flutter, Firebase, Gemini, Google Calendar OAuth 2.0, Clean Architecture, background scheduling, fallback handling, error handling, and coding standards.
- ☑ The AI usage log is retained as provided and is not modified in this revision.

Course note: AI-assisted drafting should be verified against real stakeholder feedback, and every FR should be understandable and explainable by the team during evaluation.